Last update : 2026-02-14

Benjamin Côté

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Linkedin Scholar Personal website

GRADUATE STUDIES

Ph. D. in Actuarial Science 2024 to ------
University of Waterloo. Advisors: Prof. Ruodu Wang and Prof. Alexander Schied

M. Sc. in Actuarial Science 2022 to 2024
Université Laval. Advisors: Prof. Etienne Marceau and Prof. Hélène Cossette.

UNDERGRADUATE STUDIES

Microprogram in Greek Studies 2022 to 2023
Université Laval

B. Sc. in Actuarial Science 2019 to 2022
Université Laval

Sciences, lettres et arts *(Sciences, Literature and Arts integrated program)* 2017 to 2019
Cégep de Sainte-Foy

PROFESSIONAL TITLES

Associate of the Canadian Institute of Actuaries (**ACIA**) 2025

PUBLICATIONS

Published:

Cossette, H., Côté, B., Dubeau, A. and Marceau, E. (2026). On a risk model with tree-structured Poisson Markov random field frequency, with application to rainfall events. *ASTIN Bulletin*, *forthcoming.*

Côté, B., Cossette, H. and Marceau, E. (2025). Tree-structured Markov random fields with Poisson marginal distributions. *Journal of Multivariate Analysis*, **207**, 105418.

Submitted:

Côté, B., Wang, R. and Wu, Q. (2025). Risk-insurance parity. *arXiv:2512.09208*

Côté, B., Cossette, H. and Marceau, E. (2025). Tree-structured Ising models under mean parameterization. *arXiv:2507.18749*

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Côté, B. and Wang, R. (2025). On convex order and supermodular order without finite mean. *arXiv:2502.17803*.

Côté, B., Cossette, H. and Marceau, E. (2024). Centrality and shape-related comparisons in a tree-structured Markov random field. *arXiv:2410.20240*.

PEER-REVIEW SERVICES - Total : 4 research papers

ASTIN Bulletin	since 2024
Journal of Mathematical Economics	since 2025
Mathematics of Operations Research	since 2025

TEACHING

ACTSC-431 Casualty and Health Insurance Mathematics 2. University of Waterloo.	W2026

TEACHING ASSISTANCE

ACTSC-614: Corporate Finance and Accounting. University of Waterloo.	F2025
ACTSC-611: Financial Mathematics 1. University of Waterloo.	F2025
ACTSC-625 : Casualty & Health Insurance Mathematics. University of Waterloo.	W2025
ACTSC-624 : Stochastic Processes for Actuarial Science. University of Waterloo.	W2025
ACTSC-231 : Introductory Financial Mathematics. University of Waterloo.	F2024
MTHEL-131 : Introduction to Actuarial Practice. University of Waterloo.	F2024
ACT-3000 : Risk Theory. Université Laval.	F2022 and F2023
ACT-2001 : Introduction to Actuarial Science II. Université Laval.	W2022
ACT-2005 : Non-Life Mathematics I. Université Laval.	F2021
ACT-1003 : Complements of Mathematics. Université Laval.	F2020 and F2021

PRESENTATIONS

(2026) Best- and worst-case effectiveness of monotone-response treatments. 1st ASTIN Bulletin Conference. Zurich, Switzerland. (Joint work with Ruodu Wang.)

(2025) Centrality and topology properties in a tree-structured Markov random field. Montreal-Guanajuato Workshop on Probability and Machine Learning. Guanajuato City, Guanajuato, Mexico. (Joint work with Hélène Cossette and Etienne Marceau.)

(2024) Centrality and topology properties in a tree-structured Markov random field. Waterloo Student Conference in Statistics, Actuarial Science and Finance. Waterloo, ON. (Joint work with Hélène Cossette and Etienne Marceau.)

(2024) Tree-based Markov random fields with Poisson marginal distributions. Insurance : Mathematics and Economics Conference, Chicago, IL. (Joint work with Hélène Cossette and Etienne Marceau.)

(2024) Tree-based Markov random fields with Poisson marginal distributions. Quantact Day. Montréal, QC. (Joint work with Hélène Cossette and Etienne Marceau.)

(2023) Risk models defined on a family of tree-based Markov random fields with Poisson marginals. Waterloo Student Conference in Statistics, Actuarial Science and Finance. Waterloo, ON. (Joint work with Hélène Cossette and Etienne Marceau.)

(2023) Risk models defined on a family of tree-based Markov random fields with Poisson marginals. Actuarial Research Conference. Des Moines, IA. (Joint work with Hélène Cossette and Etienne Marceau.)

(2023) Modèles de risques définis sur une famille de champs aléatoires markoviens arborescents avec marginales Poisson. Séminaire d'été étudiant du Centre Interdisciplinaire de Modélisation Mathématique de l'Université Laval. Québec, QC. (Joint work with Hélène Cossette and Etienne Marceau.)

SCHOLARSHIPS AND AWARDS

Comprehensive Exam Scholarship	2025
Russell Hiscock Graduate Scholarship in Statistics and Actuarial Science	2025
NSERC Postgraduate Scholarships – Doctoral	2024 to 2027
Bourse de formation au doctorat – FRQNT (declined)	2024 to 2028
David A. Sprott Entrance Scholarship	2024
IÉAM Excellence award in Ancient Greek language	2024
Canada Graduate Scholarships – Master's Program - NSERC	2023-2024
FRQNT Master's Training Scholarships - FRQNT	2023-2024
Quantact Scholarship	2022
Scholarship from the Chair of Actuarial Science of Université Laval	2022 and 2023
Intact Actuarial Science Scholarship	2021 and 2022
Morneau-Shepell Scholarship	2020
Governor General's Academic Medal – Bronze special	2019
Admission Scholarship in Actuarial Science	2019
Award of Excellence in Sciences, lettres et arts	2019
Clarisse-Tremblay Award in Literature	2019
Gaston-J.-Beaudoin Scholarship	2018
Lieutenant Governor's Youth Medal	2017
Lise-Leclerc Award	2017

WORK EXPERIENCE IN ACTUARIAL FIELDS

Research Assistant 2023 to 2024
Université Laval
under the supervision of Pr. H. Cossette and of Pr. E. Marceau

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Analyst in Actuarial Science 2022
iA Financial Group

Intern in Actuarial Science 2020 to 2022
iA Financial Group

OTHER WORK EXPERIENCE

Manager 2021 to 2023
Cineplex Odeon Beauport

Cast and Team Leader 2017 to 2021
Cineplex Odeon Beauport

SCRABBLE

Competitive player at the Scrabble club *Le mot passant* 2010 to 2024
Affiliated to Fédération québécoise des clubs de Scrabble francophone (FQCSF)

Participation to the World Championships of Scrabble in French 2012, 2013, 2014, 2015, 2017 and 2018
Élite, *Blitz*, *Paires* and *Défi des Jeunes* categories.

Quebec National Scrabble Champion – Junior category 2016
Quebec National Scrabble Champion – Cadet category 2013 and 2014

SOCIAL IMPLICATIONS

Member on the program comity of Sciences, lettres et arts 2022 to 2024
Cégep de Sainte-Foy